

Bang Thai Minh

Software Developer

SUMMARY

Software Developer with hands-on experience building production-grade systems and deterministic AI pipelines. Strong foundation in Node.js, Express, and Python, with practical depth in system architecture, async queues, distributed locking, and LLM integration. Experienced in translating complex requirements into maintainable, fail-safe solutions. Applies first-principles thinking to diagnose root causes and design resilient architectures.

Specialties

- Production-grade backend systems with Node.js, Express.js, TypeScript, async workers, distributed locking, graceful shutdown and zero-downtime deployment.
- Deterministic AI Engineering: LLM structured output orchestration, semantic vector search (Qdrant), MCP server development, and strict hallucination prevention architectures.
- Full-stack delivery across Angular 17, ReactJS - from API design to frontend integration.
- Data processing and validation logic for complex domain problems (logistics, medical, e-commerce).
- System thinking: translating unclear or high-level requirements into concrete, maintainable architecture.

Industry Experience

- Experience in various domains: E-Commerce Platforms, Logistics Systems, AI Applications, Game Development.

TECHNICAL STACKS

- Programming Languages: Python, JavaScript/TypeScript, SQL
- Backend Technologies & Frameworks: Node.js, Express.js, Django, Prisma, REST API, JWT, Zod
- Frontend / SPA: Angular 17, ReactJS, Tailwind CSS
- Caching & Messaging: Redis, RabbitMQ
- Database: PostgreSQL, MongoDB, SQLite
- Cloud & Infrastructure: AWS (EC2, S3, CloudFront), Docker Compose, PM2, GitHub Actions CI/CD
- Coding Principle: Clean Code, SOLID, DDD concepts, Event-driven Architecture
- Testing: Jest, Vitest, pytest
- AI / ML: Google Gemini API, Qdrant vector search, scispacy NER, Tree-sitter AST, MCP
- Methodology: Agile, Scrum

EDUCATION

- Bachelor of Engineering in Information Technology, Ho Chi Minh City University of Technology and Education (HCMUTE)

WORKING EXPERIENCE HIGHLIGHTS

FPT Software

Project: E-Commerce Platform (Mar/2026 - Jun/2026)

A production-grade full-stack e-commerce monorepo built during OJT. Delivered a highly available AWS infrastructure featuring zero-downtime deployments, AI-driven operations, and robust concurrency handling.

Role: Full-stack Developer

- System Architecture & DevOps: Architected and executed 132+ deployments to AWS EC2 using a zero-downtime PM2 cluster. Implemented automated CI/CD pipelines with auto-rollback, backup strategies, and S3/CloudFront sync.
- Frontend Optimization: Enhanced Angular 17 client performance by applying strict request throttling, debouncing, and state caching to prevent redundant API calls and race conditions during rapid product filtering and AI chatbot interactions.
- AI Integration & Performance Optimization: Developed a dual-role AI chatbot (user side for frontend navigation/orders via natural language; admin side for live dashboard analytics). Offloaded Qdrant vector sync and AI analysis to async RabbitMQ workers, slashing API latency from ~2s to <10ms.
- Payment & Concurrency: Integrated VNPay with IPN signature verification. Handled race conditions with a robust stock reservation cleanup loop and ensured idempotency for concurrent transactions.
- Security & Testing: Implemented dual rate-limiters (AI: 10/15m, Auth: 20/15m) and a Redis-backed JWT blacklist (hybrid fail-open/closed). Wrote 53+ extensive tests covering IPN signatures, stock idempotency, and Lua script atomicity.

Technologies: Node.js, Express 5, TypeScript, Prisma, Angular 17, Neon PostgreSQL, Upstash Redis, RabbitMQ, AWS EC2/S3/CloudFront, Qdrant, Google Gemini API, VNPay, Docker Compose.

Project: Movie Theater Management System (Jan/2026 - Mar/2026)

Contributed to a Django-based movie theater web application by developing custom backend modules and performance-critical features.

Role: Full-stack Developer

- Advanced Search & Data Pipeline: Engineered a hybrid search strategy utilizing PostgreSQL Trigram with GIN indexes (fuzzy matching) and B-Tree indexes (prefix search), combined with adaptive similarity thresholds. Built an internal Python ETL tool to transform unstructured JSON into normalized SQL seeds.

- Frontend Performance Optimization: Optimized ReactJS search functionality by implementing in-memory caching, debouncing, and AbortController to aggressively prevent redundant API calls and race conditions during rapid typing.
- Security & Caching: Designed a secure, time-sensitive forgot-password workflow leveraging Redis for temporary token storage. Integrated Redis caching at the API layer to significantly reduce database load.
- UI/UX & Localization: Delivered a responsive interface featuring an automated API-driven carousel and comprehensive multi-language support (i18n).

Technologies: ReactJS, Django, PostgreSQL, Redis, Mailpit, Python ETL.

Open Source / Personal R&D

Project: GraphRAG-Code (May/2026 - June/2026)

A Python-native open-source code knowledge graph that reduces AI token costs by ~90% on structural queries.

Role: Author / Solo Developer

- Built an AST-derived knowledge graph using Tree-sitter and SQLite, implementing bidirectional Personalized PageRank (PPR) via rustworkx to analyze both downstream dependencies and upstream blast-radius.
- Benchmarked retrieval performance using a custom LLM-free harness, achieving 0.98 Precision@10 on real-world repositories (outperforming the 0.27 unidirectional baseline).
- Packaged the engine as a FastMCP server, exposing zero-config context tools (get_impact, plan_change) for Cursor and Claude Desktop, including structural dead-code detection.

Technologies: Python, Tree-sitter, rustworkx, SQLite, FastMCP, Gemini API.

Project: Medical Citation Agent (May/2026 - June/2026)

A deterministic, LLM-free extraction pipeline designed for medical documents, prioritizing absolute accuracy and zero hallucination in healthcare data processing.

Role: Author / Solo Developer

- Engineered a deterministic extraction pipeline (Regex + scispacy NER) from OpenFDA SPL with exact line-level citations (document_id + start_line + raw_text) — intentionally keeping LLMs out of the extraction path to prevent medical hallucinations.
- Achieved Citation Precision 1.00 and Hallucination Rate 0.00 across 15 curated clinical test cases, backed by 96 regression tests via GitHub Actions CI.
- Designed a Rule-based SafetyGuardrail that acts as an Anti-Corruption Layer, blocking critical drug-condition pairs lacking explicit contraindication phrasing.

Technologies: Python, scispacy, regex, FastMCP, OpenFDA API, pytest.

Project: Conversational State Machine (Jun/2026)

Enterprise dialog management patterns reverse-engineered and rebuilt on an LLM-native stack.

Role: Author / Solo Developer

- Reverse-engineered the proprietary context-switching architecture of enterprise platforms (e.g., Kore.ai) through a deep-dive technical audit, then abstracted those patterns into an open, LLM-native implementation.
- Modeled four complex interruption policies (hold_and_resume, finish_then_switch, lock_current, switch_and_discard) as deterministic LIFO stack operations on a single serializable ContextObject.
- Engineered a slot-first detection pipeline using strict catalog enum constraints, enforcing deterministic structured outputs to completely eliminate entity hallucination during cross-task switches.

Technologies: Node.js, Express, TypeScript, Prisma, SQLite, Gemini API, Svelte 5.

Freelancer

Project: Container Bay Plan Validator (Sep/2025 - Dec/2025)

Develop a Python-based desktop application to support container bay plan validation in logistics operations.

Role: Python Developer

- Engineered a deterministic parser to ingest unstructured 6-digit LOC telemetry (Excel/PDF) and reconstruct it into verifiable 2D/3D maritime container stowage matrices.
- Modeled strict "Heavy-on-Light" stability algorithms and dynamic load assessments, strictly decoupling the data ingestion layer from the business rules engine.
- Parsed and interpreted 6-digit LOC codes to construct a bay/row/tier spatial model, mapping container positions into 2D UI coordinates with bay-slice navigation.
- Resolved complex structural pairing logic (virtual 40ft bays spanning dual 20ft slots) via cross-bay footprint lookups, completely eliminating false-positive validation.
- Built an interactive UI with bay-slice navigation for real-time spatial violation highlighting, packaged as a standalone executable via PyInstaller for offline port operations.

Technologies: Python, Tkinter, Pandas, PDFPlumber, Regex, PyInstaller

Project: Match-3 Puzzle Game (Jun/2025 - Sep/2025)

Developed a Match-3 puzzle game prototype based on provided visual concepts for an academic product.

Role: Game Engineer

- Architected a data-driven level engine: dynamically generated board layouts, obstacles, and win constraints entirely from JSON configurations, enabling zero-code level expansions.
- Enforced strict Event-driven architecture: utilized an Event Bus to completely decouple the GameScene (core logic) from the UIScene (rendering/HUD).
- Implemented a Board Facade pattern to orchestrate specialized, isolated modules (Matcher, Powerups, Input, State) instead of monolithic game loops.

- Modeled core mechanics (cascading behavior, tile swapping) using deterministic Finite State Machines (FSM) to prevent race conditions during rapid user inputs.

Technologies: HTML5, JavaScript, Phaser 3, Vercel